ECO 0.3.1 – 3/1/2030: (revised 4/1/2030 – kidd)

Requestor: Prof. Kidd

Description of change: Modify your design of PROJECT-MYMILLI to add a motor driving a fan with 4 possible states: OFF, LOW, MED and HIGH. The speed of the motor/fan is determined by a push button that rotates through the settings. This modification is driven by additional Customer Requirements. (See Table 1) You must implement the resulting new High-level Requirements (see Table 2) by making changes to the Design Documents you created for PROJECT-MYMILLI.

Reason for change: The customer has added the new requirement of having a cooling fan added to the device.

DEFINITIONS:

“arduino.h” – helper functions for the Arduino development platform maintained by Arduino AG, owned by Qualcomm.
CFM – Cubic Feet per Minute
The Customer – the entity/person that has contracted for the design and prototype of The Product.
Customer Requirements – Product requirements provided by the Customer
Design Documents – The formal Engineering Design and Testing documents that embody the design of the Product
The Evaluator – Entity/person evaluating whether the design and prototype satisfy the High-level requirements
High-level Technical Requirements – Direct translation of the Customer Requirements into engineering language that can be implemented
IP – Intellectual Property
ISR – Interrupt Service Routine
LED – Light Emitting Diode
The Product/System – The device being designed and prototyped
PROJECT-MYMILLI – The previous project upon which this project is based
PWM – Pulse Width Modulation
RPM – Revolutions Per Minute

CUSTOMER REQUIREMENTS: Changed, see Table 1.
HIGH-LEVEL REQUIREMENTS: Changed, see Table 2.
EDUCATIONAL CONSTRAINTS: Changed, see Table 3.

EC 1.2 The Evaluator must be able to successfully have The System perform the dialog in Table 4 and Table 5. (added – kidd, 3/20/2030)
EC 1.3 H.16, H.17, H.18 do not have to be explicitly measured. Just provide some evidence that these requirements have been satisfied though justified inference such as through calculations.